



ESOFT UNIVERSITY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)

CONTINUOUS ASSESSMENT – 2026

CCU1304 – FUNDAMENTALS OF PROGRAMMING

Assessment Type: Portfolio | Weighting: 40%

ASSESSMENT TITLE

QuickBite – Design, Implement and Test a Procedural Programming Solution

ASSESSMENT OVERVIEW

You are required to design, implement, document, and test a console-based programming solution for the QuickBite food ordering system. The assessment evaluates your ability to analyse a problem, design an algorithm, implement a procedural program, apply programming concepts, and test the completed solution.

LEARNING OUTCOMES ASSESSED

- LO1: Demonstrate awareness of the principles of programming.
- LO2: Design programming solutions.
- LO3: Implement procedural programming solutions.
- LO4: Test procedural programming solutions.

SCENARIO

QuickBite is a food delivery service that requires a computer program to calculate the total cost of customer orders. The system should accept customer information, calculate the food cost, determine the delivery charge, apply any applicable discount, validate input data, and display a clear itemised bill.

PROGRAM REQUIREMENTS

The program must accept the following inputs:

- Customer name
- Order type: Regular, Premium, or VIP
- Number of food items
- Price per item
- Delivery distance in kilometres
- Discount eligibility

The program must apply the following business rules:

- Delivery charge: Rs. 50 per kilometre.
- Premium customers receive a 10% discount on the food subtotal.
- VIP customers receive a 20% discount on the food subtotal.
- Regular customers do not receive an automatic order-type discount.
- Maximum delivery distance is 20 km.
- Orders above Rs. 5,000 qualify for free delivery.
- All monetary values must be displayed to two decimal places.
- The program must validate invalid input values.

REQUIRED SUBMISSIONS

Students must submit a single portfolio containing the following sections:

- 1. Problem Analysis:** Explain the problem, identify inputs, processing, outputs, assumptions, and constraints.
- 2. Algorithm Design:** Provide a suitable algorithm and pseudocode for the proposed solution.
- 3. Flowchart:** Create a flowchart showing the main logic of the program.
- 4. Program:** Implement the solution using the python programming language. The program should demonstrate variables, data types, operators, selection, iteration, functions, and appropriate input/output.
- 5. Input Validation:** Explain and demonstrate how invalid inputs are handled.
- 6. Testing:** Prepare a test plan and test cases covering normal, boundary, and invalid inputs.
- 7. Test Evidence:** Provide evidence of testing, including test inputs, expected outputs, actual outputs, and results.
- 8. Technical Documentation:** Explain the program structure, major functions, variables, and important programming decisions.
- 9. Reflection:** Briefly reflect on challenges faced, improvements made, and possible future enhancements.

MINIMUM TECHNICAL REQUIREMENTS

- Use meaningful variable names.
- Use appropriate data types.
- Use selection structures such as if...else or switch where appropriate.
- Use at least one loop where appropriate.
- Use at least two user-defined functions.
- Use appropriate arithmetic, relational, and logical operators.
- Include comments to explain important sections of the code.
- Validate important user inputs.
- The program should compile and execute successfully.

- The output should be clear and user-friendly.

TESTING REQUIREMENTS

Students must prepare at least 8 test cases, including:

- At least two normal/valid input cases.
- At least two boundary-value cases.
- At least two invalid-input cases.
- At least one case involving Premium order type.
- At least one case involving VIP order type.
- At least one case where free delivery applies.

CONTINUOUS ASSESSMENT MARKING SCHEME

Total: 100 marks, converted to 40% of the final module assessment.

Criterion	Description	Marks	Weight
Problem Analysis	Clear understanding of the scenario, requirements, inputs, outputs, assumptions, and constraints.	10	10%
Algorithm and Pseudocode	Logical, complete, clear, and appropriate solution design.	15	15%
Flowchart	Correct symbols, sequence, decisions, loops, and logical flow.	10	10%
Program Implementation	Correct C syntax, functionality, calculations, control structures, functions, and program structure.	25	25%
Input Validation	Appropriate validation and handling of invalid or unexpected inputs.	10	10%
Testing and Test Evidence	Test plan, valid/boundary/invalid cases, expected vs actual results, and evidence.	15	15%
Technical Documentation	Clear explanation of program structure, variables, functions, and design decisions.	10	10%
Code Quality and Presentation	Readable code, meaningful names, comments, formatting, and professional presentation.	5	5%
TOTAL		100	100%

DETAILED MARKING GUIDELINES

1. Problem Analysis – 10 Marks

- Clear explanation of the QuickBite problem – 2 marks.
- Correct identification of inputs – 2 marks.
- Correct identification of processing/calculations – 2 marks.
- Correct identification of outputs – 2 marks.
- Relevant assumptions and constraints – 2 marks.

2. Algorithm and Pseudocode – 15 Marks

- Correct sequence of steps – 4 marks.
- Correct decision logic for order types and discounts – 4 marks.
- Correct delivery charge and free-delivery logic – 2 marks.
- Correct validation logic – 2 marks.
- Clear, complete, and understandable pseudocode – 3 marks.

3. Flowchart – 10 Marks

- Correct use of standard flowchart symbols – 2 marks.
- Correct sequence and data flow – 3 marks.
- Correct decision and branching logic – 3 marks.
- Clear presentation and completeness – 2 marks.

4. Program Implementation – 25 Marks

- Correct input and variable declarations – 3 marks.
- Correct calculations – 5 marks.
- Correct use of selection structures – 4 marks.
- Correct use of iteration where appropriate – 3 marks.
- Appropriate use of user-defined functions – 4 marks.
- Correct output and formatting – 3 marks.
- Program compiles and executes correctly – 3 marks.

5. Input Validation – 10 Marks

- Validation of numeric values and non-negative quantities – 3 marks.
- Validation of order type – 2 marks.
- Validation of delivery distance – 2 marks.
- Appropriate handling and display of errors – 3 marks.

6. Testing and Test Evidence – 15 Marks

- Test plan – 3 marks.
- Normal valid test cases – 3 marks.
- Boundary-value test cases – 3 marks.
- Invalid-input test cases – 3 marks.
- Expected results, actual results, and evidence – 3 marks.

7. Technical Documentation – 10 Marks

- Explanation of program structure – 3 marks.

- Explanation of variables and data types – 2 marks.
- Explanation of functions and logic – 3 marks.
- Explanation of design decisions – 2 marks.

8. Code Quality and Presentation – 5 Marks

- Readable and well-formatted code – 2 marks.
- Meaningful names and appropriate comments – 2 marks.
- Professional portfolio presentation – 1 mark.

SUGGESTED TEST CASE FORMAT

Test ID	Input Description	Expected Result	Actual Result	Pass/Fail	Evidence
TC01	Valid Regular order	Correct total calculated			
TC02	Valid Premium order	10% discount applied			
TC03	Valid VIP order	20% discount applied			
TC04	Order above Rs. 5,000	Free delivery applied			
TC05	Delivery distance = 20 km	Accepted			
TC06	Delivery distance > 20 km	Validation error			
TC07	Negative number of items	Validation error			
TC08	Invalid order type	Validation error			

SUBMISSION REQUIREMENTS

- Submit the portfolio according to the deadline provided by the lecturer.
- The submitted program must be the student's own work.
- All external sources must be acknowledged using appropriate references.
- The source code must be included in the portfolio.
- Screenshots or other evidence should clearly demonstrate testing.
- The final portfolio should be professionally formatted.

ACADEMIC INTEGRITY

Students are expected to submit their own work. Any use of external sources, code examples, or other materials must be appropriately acknowledged. Unauthorised copying or plagiarism may be subject to the applicable academic regulations.

END OF CONTINUOUS ASSESSMENT BRIEF